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Weight-Optimized Ensemble Deep Learning for Accurate Mammographic Image Classification

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Abstract. Early diagnosis and treatment of cancer play a crucial role in improving patient outcomes and quality of life. In recent years, deep learning has significantly advanced medical imaging analysis, supporting clinicians in achieving more accurate and efficient diagnoses. However, individual deep learning models often exhibit sensitivity to variations in input data, potentially leading to inconsistent predictions and degraded classification performance. In this paper, we propose a robust ensemble framework that integrates multiple deep learning models and optimizes their contribution weights to minimize classification errors. Prior to training, the input data undergo comprehensive preprocessing, including noise reduction, normalization, and image enhancement, to ensure high-quality and consistent representations. By systematically optimizing model weights within the ensemble, the proposed approach achieves superior stability and improved generalization capability. Experimental results demonstrate that our method attains a classification accuracy of 97.85% on the test dataset, along with notable improvements in precision, recall, and F1-score. The findings highlight the potential of the proposed ensemble-based strategy to enhance the reliability of early breast cancer detection systems and promote the effective integration of advanced information technologies into medical diagnostics. Ultimately, this research contributes to better clinical decision support and fosters more timely and accurate patient care.

Keywords: Breast cancer · Mammogram · Ensemble deep learning

Anh Cang Phan, Long Hien Ho and Thuong Cang Phan: These authors contributed equally to this work.

1 Introduction

In females, breast cancer demonstrates a markedly increased incidence and fatality rate. Mammography is crucial for the prompt detection of abnormalities, enabling efficient therapy. This method is utilized for regular screening in women exhibiting concerning clinical indicators. Recent improvements in deep learning models have demonstrated efficacy in medical picture classification. The amalgamation of many models improves precision and reduces inaccuracies in the categorization procedure. Lima et al. [7] proposed a multi-resolution wavelet decomposition of mammographic ROIs followed by Zernike moment extraction and classification using SVM and Extreme Learning Machines with modified kernels. On 355 fatty-breast images from the IRMA database (233 normal, 72 benign, 83 malignant), the best model achieved 94.11% accuracy. The method combines texture and shape features in a single feature vector and is notably 50 times more efficient in accuracy-per-training-time ratio than the best prior art. Limitations include evaluation on fatty breasts only, relatively small dataset, and lack of comparison against modern deep-learning baselines. Shen et al. [4] introduced an end-to-end all-convolutional deep-learning model trained with both whole-image labels and lesion-level annotations in a two-stage approach. On the CBIS-DDSM dataset the four-model ensemble reached AUC 0.91 (digitized film) and on INbreast full-field digital mammograms AUC 0.98 (sensitivity 86.7%, specificity 96.1%). The model transfers successfully from digitized to digital mammograms with minimal fine-tuning and does not require lesion annotations after the initial stage. This is one of the earliest works demonstrating that pure deep-learning systems can outperform traditional CAD and approach radiologist-level performance on public datasets. Novikov et al. [8] proposed InvertedNet, a fully convolutional architecture with delayed subsampling, exponential linear units, and Dice-based loss for multi-class segmentation of lungs, heart, and clavicles in chest radiographs (JSRT/SCR dataset, 247 images). The model achieved mean Jaccard indices of 95.0% (lungs), 88.2% (heart), 86.8% (clavicles), outperforming U-Net by a large margin despite having ten times fewer parameters and surpassing human observer performance on lungs and heart. The architecture elegantly handles severe class imbalance and delivers state-of-the-art results on a classic benchmark with very lightweight design. Wu et al. [5] trained a very high-capacity patch-level + breast-level deep convolutional network on 200 000 screening exams (1 million images) with biopsy-proven labels. The final model achieved AUC 0.895 for cancer detection on a non-enriched test set. In a reader study with 14 radiologists (720 exams each), the AI model matched average radiologist performance, and a hybrid radiologist+AI model significantly outperformed radiologists alone. This large-scale study remains one of the strongest clinical validations showing that deep learning not only matches but can improve practicing radiologists when used as a second reader. Stelzer et al. [3] combined MaZda texture features from magnification views of BI-RADS 4 calcifications with a multilayer perceptron neural network. In 226 patients (150 benign, 76 malignant) the model yielded AUC 0.82-0.83 and, at a high-sensitivity operating point, could potentially avoid 37–46% of unnecessary benign biopsies at

the cost of only one false-negative per reader. The study shows that relatively simple radiomics + classic machine-learning approaches can still provide clinically meaningful risk stratification for the specific subgroup of suspicious calcifications. Ribli et al. [6] applied Faster R-CNN, originally an object-detection framework, to simultaneously detect and classify lesions in screening mammograms. The method achieved AUC 0.95 on INbreast and AUC 0.85 (2nd place) in the Digital Mammography DREAM Challenge. At high sensitivity it produced very few false-positive marks per image. This was one of the first works to demonstrate that a modern object-detection architecture can be directly and successfully applied to mammographic CAD, outperforming previous specialized systems while remaining fully automatic and end-to-end trainable. Based on the problems of previous research works, it can be seen that the effectiveness and role of machine learning models in improving the quality of treatment and early diagnosis of cancer on mammograms. Therefore, to lay a solid foundation for the approach proposed in this study, the next section will focus on synthesizing and analyzing Related Work, before moving on to describe the Methodology in detail in Sect. 3, Sect. 4 presents the Experimental Results, and finally discuss the main conclusions in Conclusion presented in Sect. 5.

2 Related Work

2.1 Mammography in Breast Cancer Screening

Breast cancer is a malignant disease originating from mammary gland tissue, currently the most common cancer in women worldwide and also the leading cause of cancer death in women. The disease occurs when cells in the mammary gland proliferate abnormally, lose control and form malignant tumors; if not detected and treated, these tumors can invade nearby tissues or metastasize to other organs in the body, threatening the patient's life. In the early stages, breast cancer often does not cause obvious symptoms, so early screening and detection is important in improving the prognosis for patients. Mammography [1] is an imaging technique that uses low-dose X-rays to take pictures of the breasts, allowing the internal structure of breast tissue to be observed on X-ray films. Figure 1 shows two positions, CC and MLO, that are widely used in the diagnosis of mammographic abnormalities. Mammography plays an important role in the early detection and diagnosis of breast cancer: this method can help detect abnormal changes in the breasts very early - before a woman has symptoms or feels a lump.

2.2 Deep Learning

ResNet50V2 [9] is a notable Residual Network (ResNet) variant that addresses gradient vanishing in deep networks by skip connections. ResNet152, with 152 layers, uses residual blocks (1×1 , 3×3 , 1×1) to reduce data dimensionality before 3×3 convolution, preserving memory. Due to its ability to learn deep hierarchical features, ResNet152 is suitable for large datasets like ImageNet. The VGG19

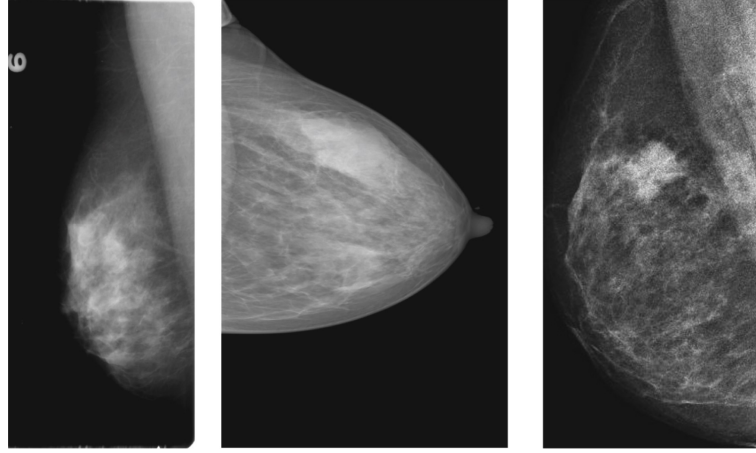


Fig. 1. Mammography images in CC and MLO views

[10] architecture, with 19 layers, features a simple design with 3×3 convolutional blocks and a 2×2 Max Pooling layer. The main advantage of this network is its consistency, gained through the use of 3×3 filters. This allows the network to learn local properties at each level, such as edges, textures, and complex objects. Due to its 144 million parameters, the VGG19 network is stable but requires a lot of computing power. Segmentation and style transfer often use VGG19 as a pre-trained network. Xception [11] enhances Inception by entirely substituting traditional convolution with Depthwise Separable Convolution. This approach enables the network to apply filters to each channel independently, substantially decreasing parameters relative to conventional convolution while preserving high classification performance.

2.3 Ensemble Learning

Ensemble Deep Learning [12] enhances classification accuracy and stability through the integration of multiple deep learning models. The ensemble of models diminishes variance, mitigates overfitting, and enhances generalization. Hard and soft voting are prevalent methods for aggregating model outputs. Examine a set of machine learning models $t_1, t_2, \dots, t_i$. Each model t_i predicts for sample x either a label $t_i(x)$ from the set $0, 1$ or a probability $p_i(c|x)$, where c is an element of $0, 1$. Each model has an associated weight $z_i \geq 0$, indicating the confidence of its predictions on the validation set. The purpose of soft voting and hard voting mechanisms is to determine the final predicted label l for sample x by synthesizing multiple machine learning models.

Hard Voting: Hard voting [13] represents a simple yet widely adopted ensemble strategy, particularly prevalent in classification problems. In this scheme, each individual classifier independently assigns a label (benign or malignant) to the input image, and the final prediction is obtained through majority rule. In other words, the class receiving the greatest number of votes becomes the ensemble's

output label.

$$\hat{l}_{\text{hard}} = \text{mode}(\{t_1(x), t_2(x), \dots, t_i(x)\}) \quad (1)$$

For the weighted variant of hard voting, the contribution of each model is scaled according to an assigned weight:

$$\hat{l}_{\text{whard}} = \arg \max_c \left(\sum_{i=1}^k z_i \cdot 1_{\{t_i(x)=c\}} \right) \quad (2)$$

Here, z_i denotes the weight associated with the i th classifier, and $1_{\{m_i(x)=c\}}$ equals 1 if model i predicts class c , and 0 otherwise.

Soft Voting: Soft voting [13] is a more refined ensemble mechanism, used when base classifiers output probability estimates rather than hard class labels. Instead of counting votes, this method aggregates the predicted probabilities from all constituent models. The final label is the class with the highest averaged (or weighted) probability. The general formulations of soft and weighted soft voting are expressed as:

$$\hat{l}_{\text{soft}} = \arg \max_c \left(\frac{1}{k} \sum_{i=1}^k p_i(c|x) \right) \quad (3)$$

$$\hat{l}_{\text{wsoft}} = \arg \max_c \left(\sum_{i=1}^k z_i \cdot p_i(c|x) \right) \quad (4)$$

Where n is the total number of classifiers, and $p_i(c|x)$ represents the probability produced by model i that the input x belongs to class c .

2.4 Evaluation Metrics

To evaluate the model's effectiveness in detecting and classifying breast abnormalities in mammograms, standard machine learning metrics are utilized. These include accuracy on the entire test set, sensitivity, precision, F1-score, and the loss value monitored during training. The metrics are computed according to the following equations:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

$$\text{Loss}(L, P) = - \sum_i L_i \log(P_i) \quad (6)$$

$$mAP = \frac{1}{n} \sum_{i=1}^n AP_i \quad (7)$$

Where TP refers to the number of benign samples correctly identified, TN the number of malignant samples correctly identified, FP the benign samples misclassified as malignant, and FN the malignant samples misclassified as benign. L_i denotes the true label for class i , P_i is the predicted probability for class i , n is the number of classes, and AP_i is the average precision for class i .

3 Methodology

In this study, three heterogeneous convolutional neural network backbones ResNet50V2, VGG19, and Xception are jointly employed to exploit their complementary architectural strengths and diverse feature learning capabilities. VGG19 adopts a deep and uniform sequential design, which is effective in capturing stable and fine-grained textural patterns of breast tissue. ResNet50V2 incorporates residual connections that facilitate the learning of deeper and more discriminative representations while alleviating gradient degradation, a property particularly beneficial for medical images that are often affected by noise and low contrast. In contrast, Xception is built upon depthwise-separable convolutions, enabling efficient modeling of spatial channel correlations with reduced computational complexity and enhanced feature compactness. The integration of these architecturally diverse models allows the ensemble to capture multiple and complementary perspectives of mammographic image characteristics, rather than relying on a single feature extraction mechanism. This diversity is crucial for reducing model bias and variance, thereby improving robustness and generalization. The effectiveness of combining these backbones is further supported by the weight correlation and importance analyses, which reveal that each model contributes differently to the final ensemble decision. Consequently, the proposed framework benefits from a synergistic combination of deep, residual, and lightweight convolutional representations. The overall methodology consists of three main stages: (1) image preprocessing and data preparation, (2) independent feature extraction and training of the backbone models, and (3) ensemble weight optimization and final classification, as illustrated in Fig. 2.

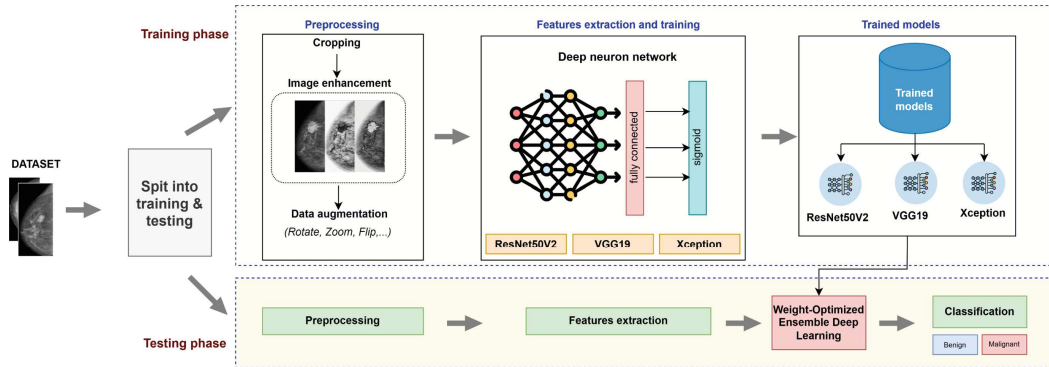


Fig. 2. Proposed general model for weight-optimizing ensemble deep learning

3.1 Training Stage

During the training phase, the input images undergo several preprocessing steps, including cropping, image enhancement, and various data augmentation techniques such as flipping, rotation, and zooming. After preprocessing, the images

are fed into the selected deep learning architectures to enable feature learning and extraction. The dataset is trained separately on three convolutional neural network models VGG19, ResNet50V2, and Xception using identical training settings and hyperparameters. Each model is optimized independently. To prevent overfitting, an early stopping strategy is applied throughout the training process. For performance assessment, we retain the best-performing version of each model based on its training outcomes.

3.2 Testing Stage

The test image undergoes data preprocessing, followed by feature extraction, after which hard voting and soft voting approaches, along with a weight optimization method applied to three models, yield the final classification as benign or malignant.

3.3 Weight-Optimization Algorithm

The weight optimization procedure is conducted exclusively on the validation set to prevent data leakage and to ensure an unbiased evaluation. All datasets are split at the patient level, guaranteeing that images from the same patient appear in only one subset training, validation, or test throughout the entire experimental process. After independently training the backbone models, their prediction scores on the validation set are collected and used as inputs for the ensemble optimization stage. The optimization objective is to identify a set of model weights that maximizes classification accuracy. To achieve this, a random search strategy is employed. For each ensemble configuration, 100 optimization trials are performed. In each trial, model weights are randomly sampled from a uniform distribution in the range $[0, 1]$ and subsequently normalized so that their sum equals one. The proposed algorithm (Algorithm 1) is an ensemble learning method that uses weight optimization to improve image classification performance.

Depending on the selected ensemble scheme, either weighted hard voting or weighted soft voting is applied to aggregate the predictions of the base models. The resulting ensemble predictions are then compared with the corresponding ground-truth labels in the validation set, and the achieved accuracy is recorded. Across trials, the weight configuration that yields the highest validation accuracy is selected as the optimal solution. Empirical analysis shows that validation performance converges after a hundred trials, indicating that random search sufficiently explores the low-dimensional weight space defined by the limited number of ensemble members. Random search is therefore well suited for this task, offering an efficient balance between exploration capability and computational cost. In contrast, grid search becomes impractical even at moderate resolutions, while Bayesian optimization introduces additional modeling complexity that provides limited benefit in such a smooth and low-dimensional optimization landscape. Consequently, random search represents a practical and effective choice for optimizing ensemble weights in the proposed framework (Table 1).

Algorithm 1 Weight-Optimized Ensemble Deep Learning Algorithm

```

1: Input: A set of trained models  $\{t_1, t_2, \dots, t_i\}$ ; test dataset  $X_{\text{test}}$ ; ground-truth
   labels  $y_{\text{test}}$ ; voting scheme mode  $\in \{\text{soft}, \text{hard}\}$ 
2: Retrieve the prediction scores from each model  $m_i$ 
3: Create an case study with the objective of maximizing classification accuracy
4: for each trial do
5:   Randomly assign weights  $z_1, z_2, \dots, z_m \in [0, 1]$ 
6:   Normalize weights:  $z_i \leftarrow \frac{z_i}{\sum_{j=1}^m w_j}$ 
7:   if mode = hard then
8:     Aggregate predictions using the hard voting rule (Eq. 2)
9:     Produce the predicted labels  $\hat{l}$ 
10:  else if mode = soft then
11:    Aggregate predictions using the soft weighted rule (Eq. 4)
12:    Produce the predicted labels  $\hat{l}$ 
13:  end if
14:  Evaluate  $\text{Accuracy} = \text{acc}(y_{\text{test}}, \hat{l})$ 
15:  Report  $\text{Acc}$  back to the case study
16: end for
17: Select the best-performing weight set  $z_i^*$  from the highest scoring trial
18: return predictions  $\hat{l}$  obtained with  $z_i^*$ 
19: Output: Final prediction  $\hat{l}$ , ptimal weights  $z_i^*$ 

```

Table 1. Suggested configurations and training settings for the experimental scenarios

Scenario	Model	Epochs	LR	Batch size
1	ResNet50V2	100	0.0001	32
2	VGG19	100	0.0001	32
3	Xception	100	0.0001	32
4	Ensemble Learning using Hard_Voting			
5	Ensemble Learning using Soft_Voting			
6	Weighted Ensemble (Hard_Voting Strategy)			
7	Weighted Ensemble (Soft_Voting Strategy)			

4 Evaluation Results

4.1 Runtime Environment and Evaluation Datasets

The experiments were executed in a Python-based environment on Kaggle, utilizing an Intel Xeon CPU running at 2.00 GHz, equipped with 30 GB of RAM, and supported by an Nvidia Tesla P100 GPU featuring 16 GB of dedicated memory. TensorFlow v1.5 served as the primary deep learning framework for training the models. The dataset was constructed from the VinDr repository combined with images provided by multiple hospitals. All samples were subsequently validated by medical specialists to assign benign or malignant labels. Following preprocessing where image slices containing lesions were extracted and converted into

.jpg format the final dataset comprised 6,930 mammographic images. Among them, 5,862 images formed the training set, while 1,068 images were reserved for evaluation, maintaining an 80/20 split.

4.2 Training Evaluation

This section presents the training and testing results of deep learning models utilized for the medical picture classification problem with the proposed model. We employ evaluation metrics such as Precision, Recall, F1-score, and Accuracy for each individual class to evaluate the model on the test set during the training phase. To evaluate the efficacy of each model, distinct models such as ResNet50V2, VGG19, and Xception are individually trained and assessed. According to Table 3, all three proposed models attain elevated training accuracy, with Sce.1 reaching the pinnacle at 95.1%, followed by Sce.2 at 92.3%, and Sce.3 at the lowest with 91.7%. Regarding training error, Sce.1 yields the optimal outcome with the minimal loss of 0.190, whilst Sce.3 exhibits the maximum loss of 0.246 (Table 2).

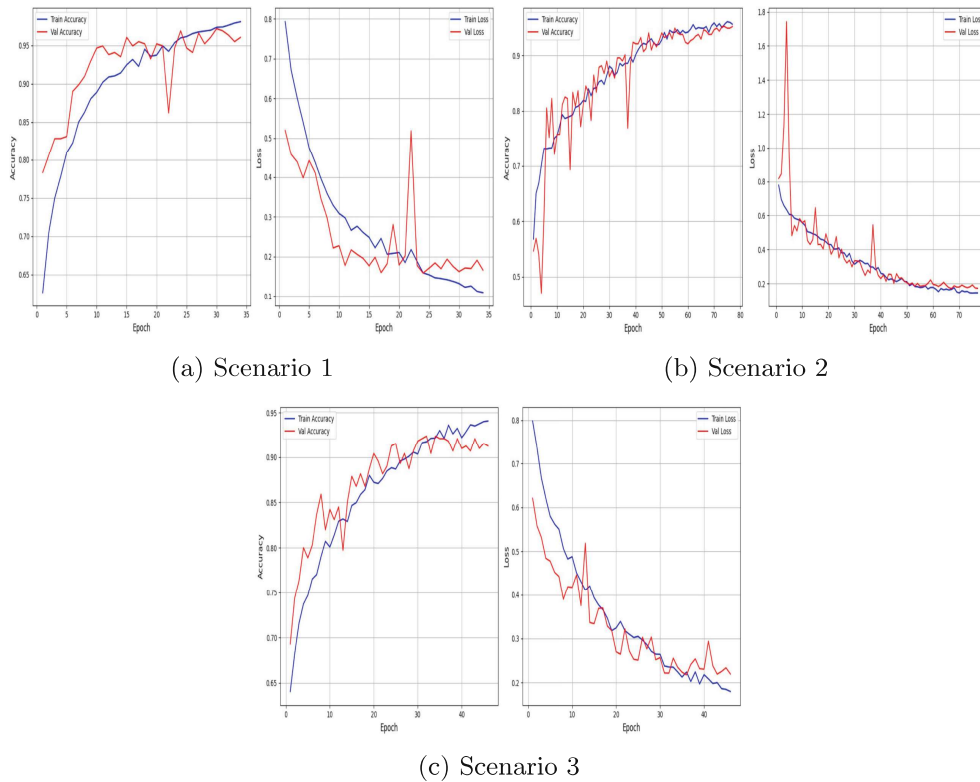
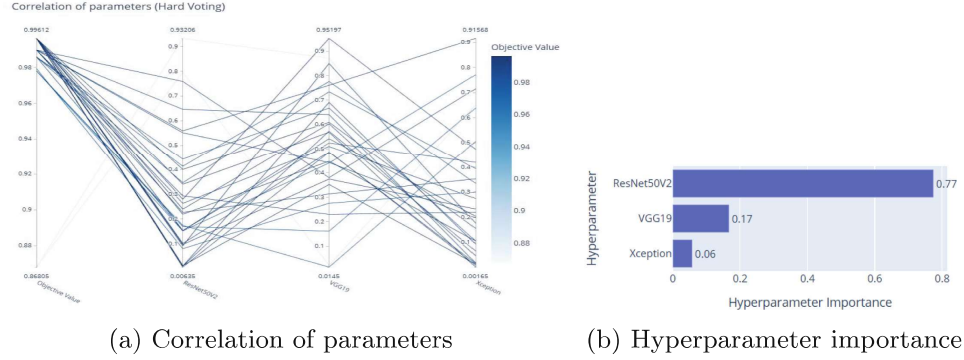


Fig. 3. Training results of 3 proposed models

Besides assessing the outcomes of each model, we are also concerned with the performance contribution of each model during the combination process.

Table 2. Training results of Sce.1,2,3.

Scenario	Accuracy (%)	Loss	Time (minutes)
Scenario 1	95.1	0.190	156.81
Scenario 2	92.3	0.232	410.25
Scenario 3	91.7	0.246	214.56

**Fig. 4.** The important weights of each model (Sce.6).

The evaluation findings encompass the correlation between the models and the significance of each model's contribution, as illustrated in Figs. 4 and 5.

Figures 4 and 5 depict the correlation between model contributions and their importance during the parameter optimization process for Sce.6 and 7. Subfigure (a) illustrates the variation of the Objective Value in relation to the given weights of the three models: ResNet50V2, VGG19, and Xception. The line colors denote the performance level of each weight configuration, with darker hues signifying superior objective results. The research indicates that superior objective values are typically attained when ResNet50V2 and VGG19 are allocated more weights, while Xception is usually assigned relatively lesser weights (Table 3).

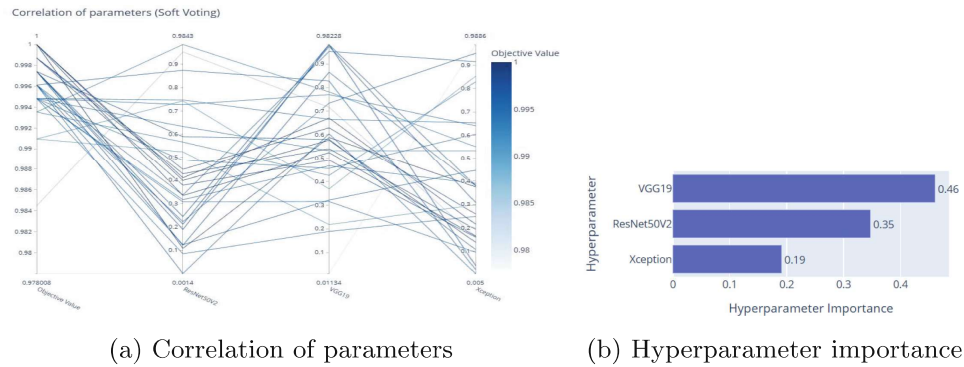
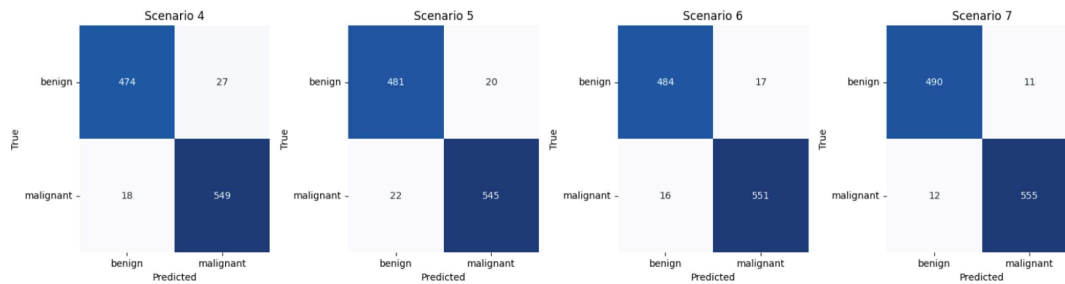
**Fig. 5.** The important weights of each model (Sce.7).

Table 3. Classification results of scenarios.

Scenario	Classify	Precision(%)	Recall(%)	F1-score(%)	mAP(%)
Scenario 1	B	95.81	94.67	95.24	95.51
	M	95.24	96.26	95.74	
Scenario 2	B	91.02	93.25	92.12	92.70
	M	94.18	92.23	93.19	
Scenario 3	B	88.82	92.32	90.54	91.29
	M	93.47	90.44	91.93	
Scenario 4	B	94.61	96.34	95.47	95.79
	M	96.83	95.31	96.06	
Scenario 5	B	96.01	95.63	95.82	96.07
	M	96.12	96.46	96.29	
Scenario 6	B	96.61	96.8	96.7	96.91
	M	97.18	97.01	97.09	
Scenario 7	B	97.8	97.61	97.71	97.85
	M	97.88	98.06	97.97	

The text in bold represents the highest accuracy across experimental scenarios

The results table presents the classification performance for the two categories, Benign (B) and Malignant (M), across seven distinct scenarios, incorporating metrics such as Precision, Recall, F1-score, and Accuracy. Sce.7 demonstrates the highest performance, achieving an Accuracy of 97.85%, with greater Precision, Recall, and F1-score in both classes (B: 97.88% Precision, M: 97.97% F1-score). This indicates that employing weight optimization is the most effective method for accurately distinguishing between the two classes (Fig. 6).

**Fig. 6.** Confusion matrix of Sce. 4, 5, 6, 7.

The confusion matrices of Sce. 4 to 7 illustrate the progressive improvement in classification performance as weighting and optimization strategies are introduced into the ensemble model. Sce.4, which applies standard hard voting without weights, still produces a noticeable number of misclassifications in

both classes benign and malignant. Sce.5, using unweighted soft voting, yields slightly better results with reduced false predictions in both categories. A more substantial improvement appears in Sce.6, where weight-optimized is applied to hard voting, leading to a marked reduction in errors, particularly for benign cases (only 16 misclassifications). Sce.7 achieves the best performance overall, with the fewest misclassified samples-only 8 benign and 15 malignant cases-demonstrating that combining weighted optimization with soft voting significantly enhances the ensemble's ability to distinguish between benign and malignant mammographic images. These results confirm that weight optimization plays a critical role in improving sensitivity and precision, especially in the final two scenarios.

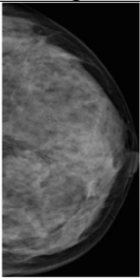
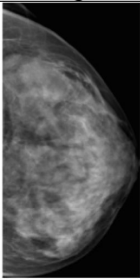
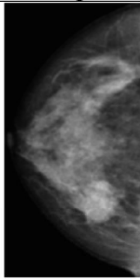
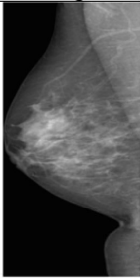
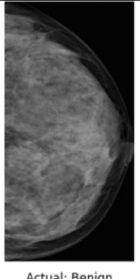
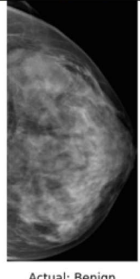
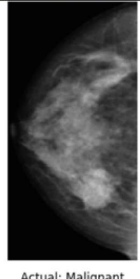
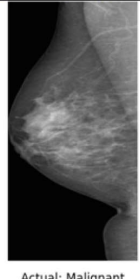
Method	Output			
	Img1	Img2	Img3	Img4
Scenario 6				
	Actual: Benign Predict: Benign Voting: 0/3 Malignant 3/3 Benign	Actual: Benign Predict: Benign Voting: 0/3 Malignant 3/3 Benign	Actual: Malignant Predict: Malignant Voting: 3/3 Malignant 0/3 Benign	Actual: Malignant Predict: Malignant Voting: 3/3 Malignant 0/3 Benign
Scenario 7				
	Actual: Benign Predict: Benign $p_{\text{soft}}(c=0): 1.00 \mid p_{\text{soft}}(c=1): 0.00$	Actual: Benign Predict: Benign $p_{\text{soft}}(c=0): 1.00 \mid p_{\text{soft}}(c=1): 0.00$	Actual: Malignant Predict: Malignant $p_{\text{soft}}(c=0): 0.00 \mid p_{\text{soft}}(c=1): 1.00$	Actual: Malignant Predict: Malignant $p_{\text{soft}}(c=0): 0.00 \mid p_{\text{soft}}(c=1): 1.00$

Fig. 7. Classification results on test dataset of Sce. 6 and 7.

Figure 7 illustrates two important scenarios in binary classification of mammograms (Benign vs. Malignant) using a set of three models with hard and soft voting. In Sce.6, based on combined hard voting mechanism and Sce.7 based on soft voting mechanism in both scenarios give good classification results on the test dataset. Based on the classification ability of the three basic models through the weight optimization process shows the efficiency in classification, limiting the confusion between classes and taking advantage of the advantages of each model contributing to the classification process.

5 Conclusion

In this study, we propose an ensemble learning algorithm to maximize the advantages of each deep learning model in the task of breast cancer classification on mammograms. The method focuses on optimizing the contribution weight of each component model, thereby minimizing the classification error and increasing the reliability of the merged model. This approach is proven effective through a system of synthetic evaluation indexes, clearly reflecting the level of improvement compared to each individual model. In addition, preprocessing and data augmentation techniques play a key role in improving the quality of the training dataset, creating diversity in image features and helping deep learning models increase their generalization ability. Processing steps such as cropping ROI regions, intensity normalization, enhancement by rotation, flipping, scaling, or contrast enhancement have contributed to significantly improving the model's feature learning ability. The experimental process was deployed on various scenarios to evaluate the stability and adaptability of the proposed method. The results show that the combined model achieved 97.85% accuracy on the test dataset, while significantly reducing the misclassification phenomenon in both groups: benign and malignant lesions. Notably, one of the core contributions of this study is to propose an algorithm to optimize the weights of the component deep learning models, which helps to determine the relative contribution of each VGG19, ResNet50V2 and Xception model to the final classification performance. This optimization not only highlights the models with stronger feature extraction capabilities but also allows to effectively take advantage of the complementarity between different CNN architectures. The results obtained confirm the feasibility, stability and high application potential of the algorithm in clinical diagnostic support systems, especially in contexts requiring high accuracy and the ability to handle complex mammographic features. In the future, we plan to expand the research by evaluating more advanced deep learning models to determine the most optimal combination. At the same time, the next development direction will focus on multi-class classification and integrating more specialized mammographic features, such as tissue density and the presence of microcalcifications, to further improve the accuracy and coverage when applied to more diverse and complex datasets.

References

1. R. S. of N. A. (RSNA) and A. C. of Radiology (ACR): Mammography. Radiology-info.org. <https://www.radiologyinfo.org/en/info/mammo>. Accessed 17 Oct 2025
2. Zonderland, H., Smithuis, R.: BI-RADS for mammography and ultrasound 2013, Updated version. Radiologyassistant.nl (2014)
3. Stelzer, P.D., Steding, O., Raudner, M.W., Euler, G., Clauser, P., Baltzer, P.A.T.: Combined texture analysis and machine learning in suspicious calcifications detected by mammography: potential to avoid unnecessary stereotactical biopsies. *Eur. J. Radiol.* **132**, 109309 (2020). <https://doi.org/10.1016/j.ejrad.2020.109309>

4. Shen, L., Margolies, L.R., Rothstein, J.H., Fluder, E., McBride, R., Sieh, W.: Deep learning to improve breast cancer detection on screening mammography. *Sci. Rep.* **9**(1), 12495 (2019). <https://doi.org/10.1038/s41598-019-48995-4>
5. Wu, N., et al.: Deep neural networks improve radiologists. In: *Performance in Breast Cancer Screening* (2019). [arXiv: arXiv:1903.08297](https://arxiv.org/abs/1903.08297). <https://doi.org/10.48550/arXiv.1903.08297>
6. Ribli, D., Horv, A., Unger, Z., Pollner, P., Csabai, I.: Detecting and classifying lesions in mammograms with deep learning (2017), [arXiv: arXiv:1707.08401](https://arxiv.org/abs/1707.08401). <https://doi.org/10.48550/arXiv.1707.08401>
7. De Lima, S.M.L., Da Silva-Filho, A.G., Dos Santos, W.P.: Detection and classification of masses in mammographic images in a multi-kernel approach. *Comput. Methods Programs Biomed.* **134**, 11–29 (2016). <https://doi.org/10.1016/j.cmpb.2016.04.029>
8. Novikov, A.A., Lenis, D., Major, D., Hladvka, J., Wimmer, M., Bhler, K.: Fully convolutional architectures for multi-class segmentation in chest radiographs (2018). [arXiv: arXiv:1701.08816](https://arxiv.org/abs/1701.08816). <https://doi.org/10.48550/arXiv.1701.08816>
9. Szegedy, C., Ioffe, S., Vanhoucke, V., Alemi, A.: Inception-v4, inception-ResNet and the impact of residual connections on learning. In: *Proceedings of AAAI Conference on Artificial Intelligence*, vol. 31, no. 1 (2017)
10. Simonyan, K., Zisserman, A.: Very deep convolutional networks for large-scale image recognition (2014). [arXiv:1409.1556](https://arxiv.org/abs/1409.1556)
11. Chollet, F.: Xception: deep learning with depthwise separable convolutions. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1251–1258 (2017)
12. Dietterich, T.G.: Ensemble methods in machine learning. In: *Proceedings of the International Workshop on Multiple Classifier Systems*, pp. 1–15 (2000)
13. Hastie, T., Tibshirani, R., Friedman, J.: *The Elements of Statistical Learning*. Springer, New York (2009)